

## SAFETY DATA SHEET

Revision Date 02-April-2024

Revision Number 4

### 1. Identification

**Product Name** Aluminum Copper spheres, alloy 2017

**Cat No. :** 42035

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Aluminum  | 7429-90-5 | 94.8     |
| Copper    | 7440-50-8 | 4.0      |
| Manganese | 7439-96-5 | 0.7      |

|           |           |     |
|-----------|-----------|-----|
| Magnesium | 7439-95-4 | 0.5 |
|-----------|-----------|-----|

#### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                   |

#### 5. Fire-fighting measures

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | approved class D extinguishers. Do not use water or foam. |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective                                  |
| <b>Flash Point</b>                      | No information available                                  |
| <b>Method -</b>                         | No information available                                  |
| <b>Autoignition Temperature</b>         | No information available                                  |
| <b>Explosion Limits</b>                 |                                                           |
| <b>Upper</b>                            | No data available                                         |
| <b>Lower</b>                            | No data available                                         |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                  |
| <b>Sensitivity to Static Discharge</b>  | No information available                                  |

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Metal oxides.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

#### 6. Accidental release measures

|                                             |                                                                                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.                               |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Pick up and transfer to properly labelled containers.                           |

#### 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage.** Keep in a dry place. Keep away from acids. Incompatible Materials. Oxidizing agent.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                                                | British Columbia                                          | Ontario TWAEV                                                                           | Quebec                                                 | ACGIH TLV                                                 | OSHA PEL                                                                                                                                            | NIOSH                                                                                 |
|-----------|--------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Aluminum  | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>  | TWA: 1.0 mg/m <sup>3</sup>                                | TWA: 1 mg/m <sup>3</sup>                                                                | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>  | TWA: 1 mg/m <sup>3</sup>                                  | (Vacated) TWA: 15 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                  | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                                 |
| Copper    | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>TWA: 0.2 mg/m <sup>3</sup>    | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                  | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup>                                | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                                      | IDLH: 100 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |
| Manganese | TWA: 0.2 mg/m <sup>3</sup>                             | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup><br>TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup>                             | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 mg/m <sup>3</sup> | IDLH: 500 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup>  |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** None under normal use conditions.

### Personal protective equipment

**Eye Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection** No special protective equipment required

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations | -               | Splash protection only |

### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do

not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                 |
|----------------------------------------|---------------------------------|
| Physical State                         | Solid Spheres                   |
| Appearance                             | Silver                          |
| Odor                                   | Odorless                        |
| Odor Threshold                         | No information available        |
| pH                                     | No information available        |
| Melting Point/Range                    | No data available               |
| Boiling Point/Range                    | No information available        |
| Flash Point                            | No information available        |
| Evaporation Rate                       | Not applicable                  |
| Flammability (solid,gas)               | No information available        |
| Flammability or explosive limits       |                                 |
| Upper                                  | No data available               |
| Lower                                  | No data available               |
| Vapor Pressure                         | <=1100 hPa @ 50 °C              |
| Vapor Density                          | Not applicable                  |
| Specific Gravity                       | No information available        |
| Solubility                             | No information available        |
| Partition coefficient; n-octanol/water | No data available               |
| Autoignition Temperature               | No information available        |
| Decomposition Temperature              | No information available        |
| Viscosity                              | Not applicable                  |
| Molecular Formula                      | Al:Cu:Mn:Mg; 94.8:4:0.7:0.5 wt% |

## 10. Stability and reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactive Hazard                  | None known, based on information available |
| Stability                        | Stable under normal conditions.            |
| Conditions to Avoid              | Incompatible products.                     |
| Incompatible Materials           | Oxidizing agent                            |
| Hazardous Decomposition Products | Metal oxides                               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.   |
| Hazardous Reactions              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Mist LC50   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |

#### Component Information

| Component | LD50 Oral                | LD50 Dermal | LC50 Inhalation               |
|-----------|--------------------------|-------------|-------------------------------|
| Aluminum  | Not listed               | Not listed  | LC50 > 0.888 mg/L ( Rat ) 4 h |
| Copper    | Not listed               | Not listed  | LC50 > 5.11 mg/L ( Rat ) 4 h  |
| Manganese | LD50 = 9 g/kg ( Rat )    | Not listed  | LC50 > 5.14 mg/L ( Rat ) 4 h  |
| Magnesium | LD50 = 230 mg/kg ( Rat ) | Not listed  | Not listed                    |

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Aluminum  | 7429-90-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Copper    | 7440-50-8 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Manganese | 7439-96-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Magnesium | 7439-95-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Algae                                                                                                                                   | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Microtox   | Water Flea                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------|
| Copper    | EC50: 0.031 - 0.054 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 0.0426 - 0.0535 mg/L, 72h static (Pseudokirchneriella subcapitata) | LC50: = 1.25 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 0.8 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.112 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: = 0.052 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.0068 - 0.0156 mg/L, 96h (Pimephales promelas)<br>LC50: < 0.3 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.2 mg/L, 96h flow-through (Pimephales promelas) | Not listed | EC50: = 0.03 mg/L, 48h Static (Daphnia magna) |

|           |            |                                                         |            |            |
|-----------|------------|---------------------------------------------------------|------------|------------|
| Manganese | Not listed | LC50: > 3.6 mg/L, 96h semi-static (Oncorhynchus mykiss) | Not listed | Not listed |
|-----------|------------|---------------------------------------------------------|------------|------------|

**Persistence and Degradability** Insoluble in water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

### 15. Regulatory information

#### International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Aluminum  | 7429-90-5 | X   | -    | X    | ACTIVE                                        | 231-072-3 | -      | -   |
| Copper    | 7440-50-8 | X   | -    | X    | ACTIVE                                        | 231-159-6 | -      | -   |
| Manganese | 7439-96-5 | X   | -    | X    | ACTIVE                                        | 231-105-1 | -      | -   |
| Magnesium | 7439-95-4 | X   | -    | X    | ACTIVE                                        | 231-104-6 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Aluminum  | 7429-90-5 | X     | KE-00881 | X    | -    | X    | X    | X     | X     |
| Copper    | 7440-50-8 | X     | KE-08896 | X    | -    | X    | X    | X     | X     |
| Manganese | 7439-96-5 | X     | KE-22999 | X    | -    | X    | X    | X     | X     |
| Magnesium | 7439-95-4 | X     | KE-22673 | X    | -    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Aluminum  | Part 1, Group A Substance                            |                                                                            |                                           |
| Copper    | Part 1, Group A Substance                            |                                                                            |                                           |

|           |                           |  |
|-----------|---------------------------|--|
| Manganese | Part 1, Group A Substance |  |
|-----------|---------------------------|--|

**Legend** NPRI - National Pollutant Release Inventory

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Aluminum  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Copper    | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Aluminum  | 7429-90-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Copper    | 7440-50-8 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Manganese | 7439-96-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Magnesium | 7439-95-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Aluminum  | 7429-90-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Copper    | 7440-50-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Manganese | 7439-96-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Magnesium | 7439-95-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

#### Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

#### Revision Date

02-April-2024

#### Print Date

02-April-2024

#### Revision Summary

New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**